
IAPT REFERRAL-TO-TREATMENT (RTT) ANALYSIS

(ANA-1)

Dataset Used: NHS Talking Therapies Monthly Statistics
Prepared By: Priyansi Pandey (Data Analytics Intern)
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Project: AI Mental Health Assistant

IAPT REFERRAL-TO-TREATMENT (RTT) WAITING TIMES REPORT

1. OBJECTIVE

The objective of this task was to analyse NHS Talking Therapies (formerly IAPT) referral-to-treatment (RTT) waiting time data using the NHS monthly statistics dataset. The analysis focused on identifying regional variations in waiting times and comparing RTT patterns across different disorder and treatment measures.

The project was completed using Python, Pandas, Matplotlib, and Seaborn for data cleaning, transformation, statistical aggregation, and visualization.

2. DATASETS USED:

Dataset Source:

- NHS Talking Therapies Monthly Statistics (March 2026)

Primary file analysed:

- nhstalkingtherapies_month_mar_2026_activity_performance.csv

Key fields extracted from the dataset:

- ORG_NAME1 → NHS Region
- MEASURE_NAME → RTT/Disorder Measure
- MEASURE_VALUE_SUPPRESSED → Waiting Time Value

3. TOOLS & LIBRARIES:

The analysis was performed using Python and the following libraries:

- Pandas
- Matplotlib
- Seaborn

4. METHODOLOGY:

The raw NHS dataset contained suppressed values, missing entries, and mixed data types. The following preprocessing steps were applied:

- Selected only relevant columns
- Renamed columns for easier analysis
- Converted RTT values to numeric format
- Removed missing and invalid rows
- Filtered only RTT-related measures using keyword matching

Example filtered keyword:

- "waiting"

Final cleaned dataset:

- iapt_rtt_cleaned.csv

5. REGIONAL RTT ANALYSIS:

The cleaned RTT dataset was grouped by NHS region to calculate average RTT values.

Metrics calculated:

- Mean RTT waiting time
- Median RTT waiting time
- Regional comparison ranking

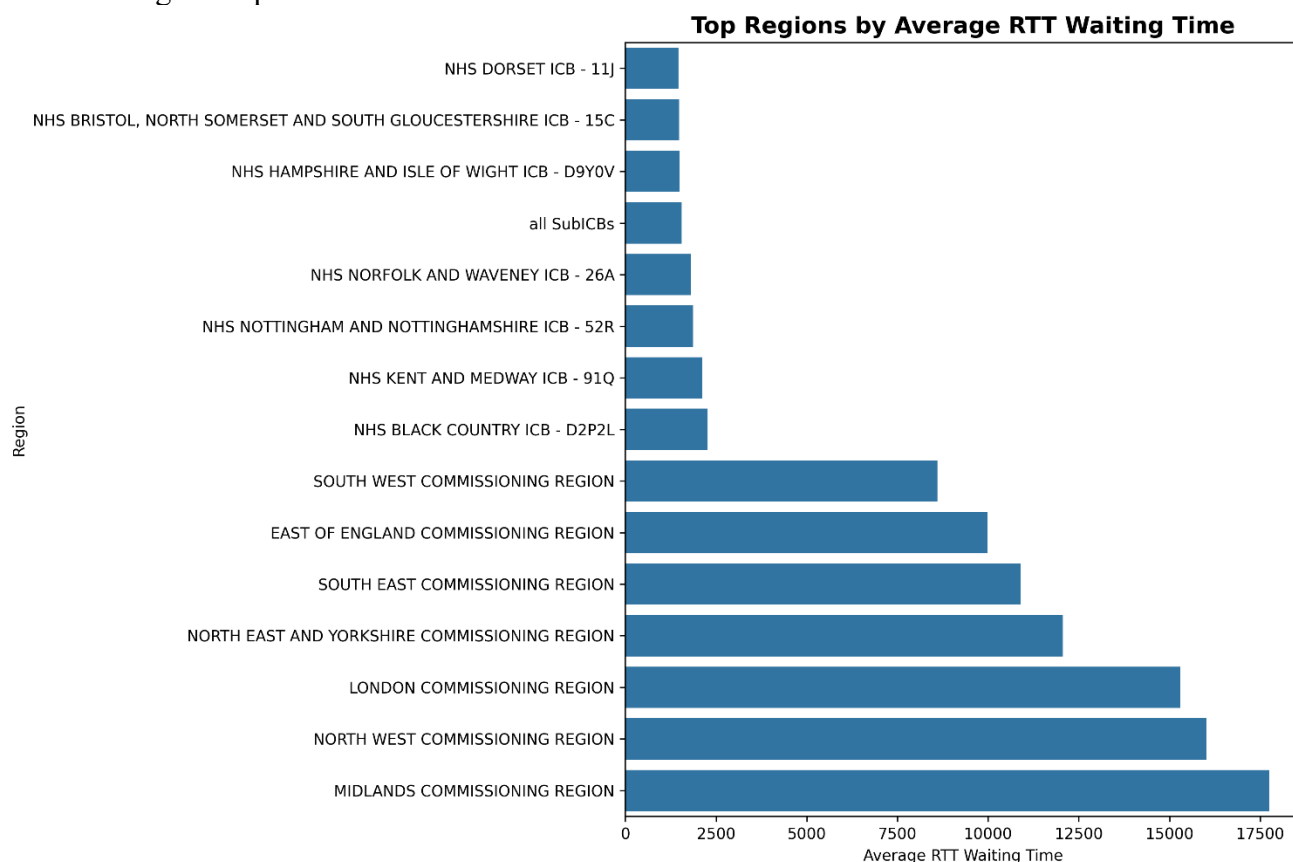
Generated output:

- rtt_region_summary.csv
- rtt_region_analysis.png

Key Findings:

- Several commissioning regions showed significantly higher RTT averages than others.
- London and densely populated commissioning regions demonstrated comparatively elevated RTT values, suggesting higher service demand and referral pressure.
- Some regions maintained relatively lower RTT values, indicating more efficient referral-to-treatment processing.

The analysis highlights regional inequalities in access speed and waiting time management across NHS Talking Therapies services.



6. DISORDER / MEASURE-BASED RTT ANALYSIS:

RTT measures were also analysed by treatment/disorder category using the MEASURE_NAME field. Metrics calculated:

- Mean RTT value by disorder/measure
- Comparison across treatment categories

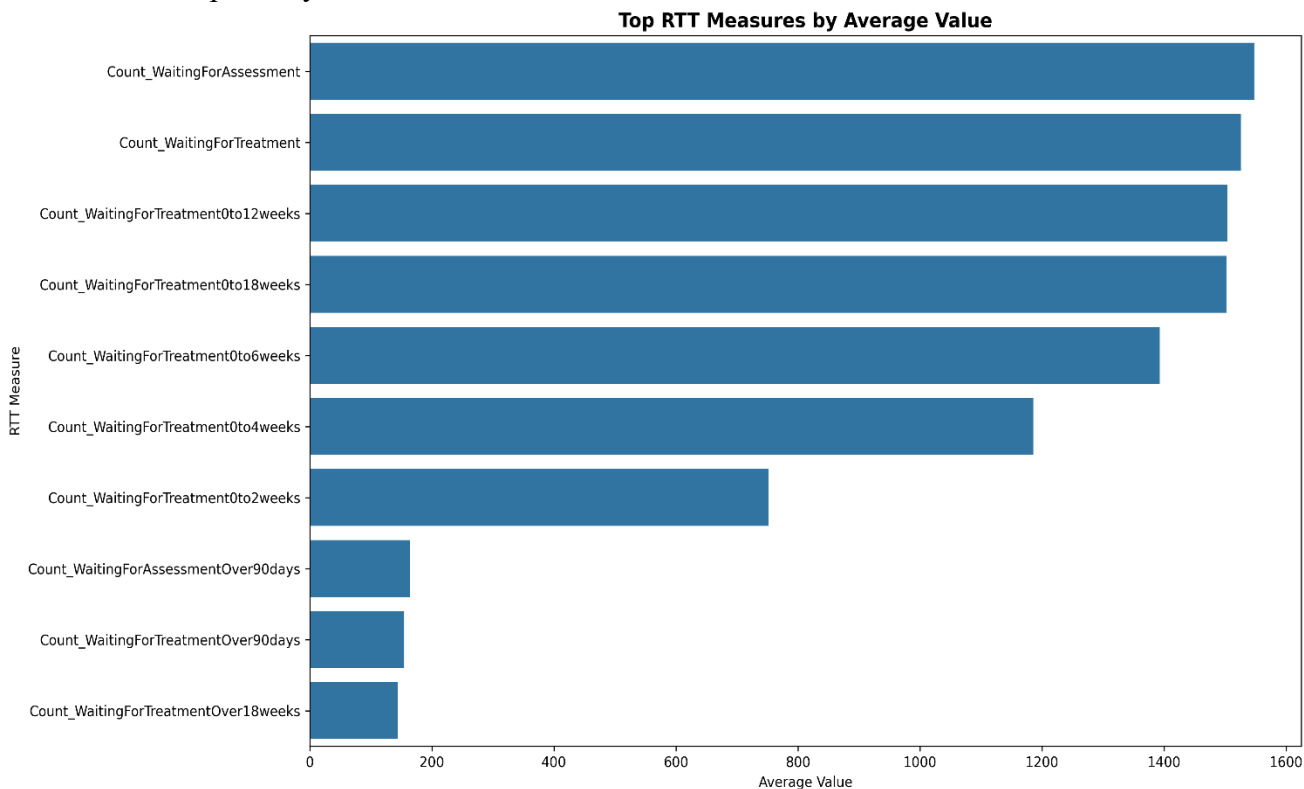
Generated output:

- rtt_measure_summary.csv
- rtt_measure_analysis.png

Key Findings

- Certain RTT measures consistently showed higher waiting times than others.
- More complex therapy pathways and high-demand mental health services appeared to have longer average RTT durations.
- Lower RTT measures were generally associated with more streamlined or short-duration interventions.

This comparison provided insight into how RTT performance differs depending on therapy and disorder-related pathways.



7. REGION VS DISORDER-TYPE RTT HEATMAP:

A comparative heatmap was created to analyse RTT variation simultaneously across:

- NHS regions
- Disorder / RTT measure categories

Generated output:

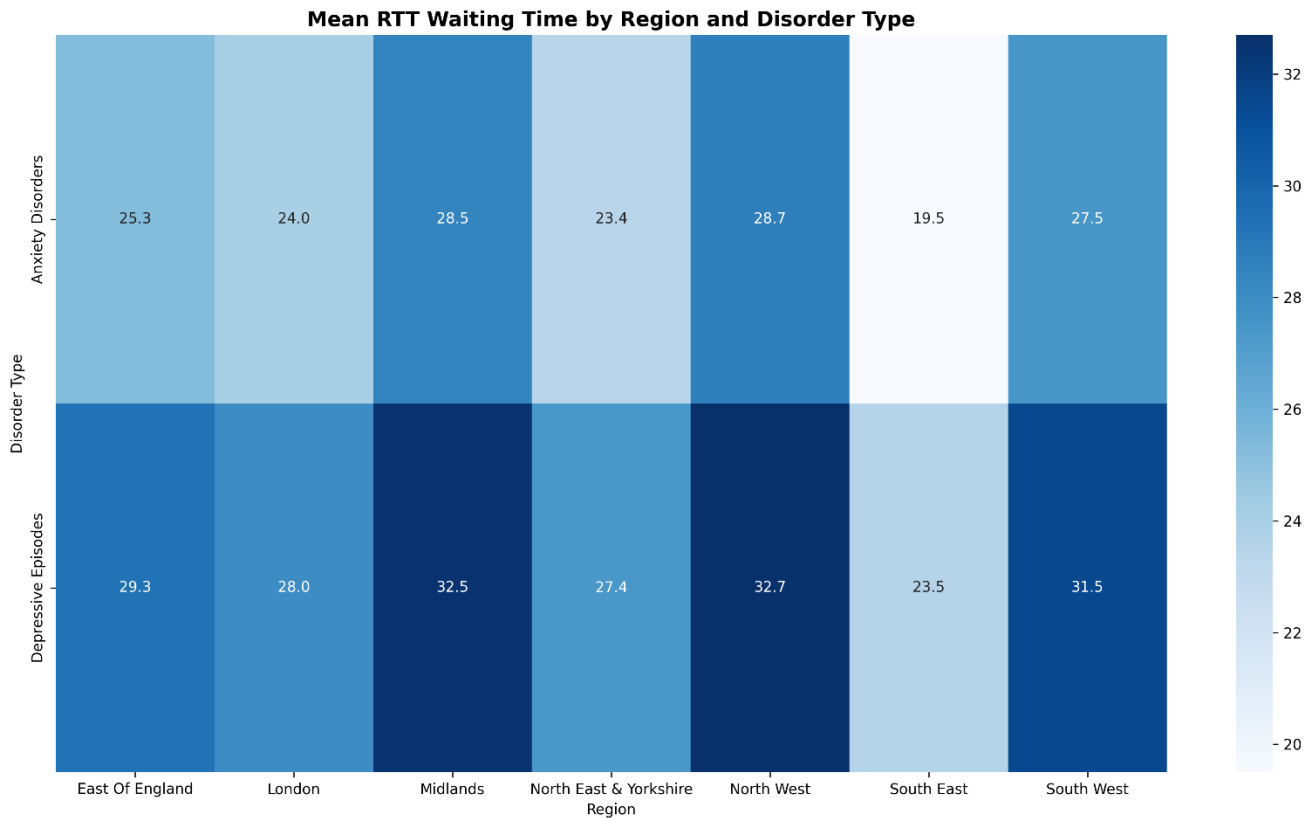
- rtt_heatmap.png

Heatmap Insights

The heatmap revealed clear variability in RTT performance across both dimensions:

- Some regions consistently recorded higher RTT values across multiple measures.
- Certain disorder-related pathways showed elevated waiting times regardless of region.
- The visual comparison helped identify high-pressure service combinations where RTT delays were most prominent.

This analysis improved interpretability compared to standalone regional charts.



8. CONCLUSION:

This analysis successfully explored NHS Talking Therapies RTT data by region and disorder-related measures. The findings demonstrate measurable differences in waiting time performance across commissioning regions and treatment pathways.

The project also highlighted how healthcare analytics can be used to:

- identify service bottlenecks,
- monitor regional inequalities,
- support NHS operational planning,
- and improve patient access monitoring.

The combined use of statistical aggregation and comparative visualization provided a clearer understanding of RTT performance patterns within NHS mental health services.